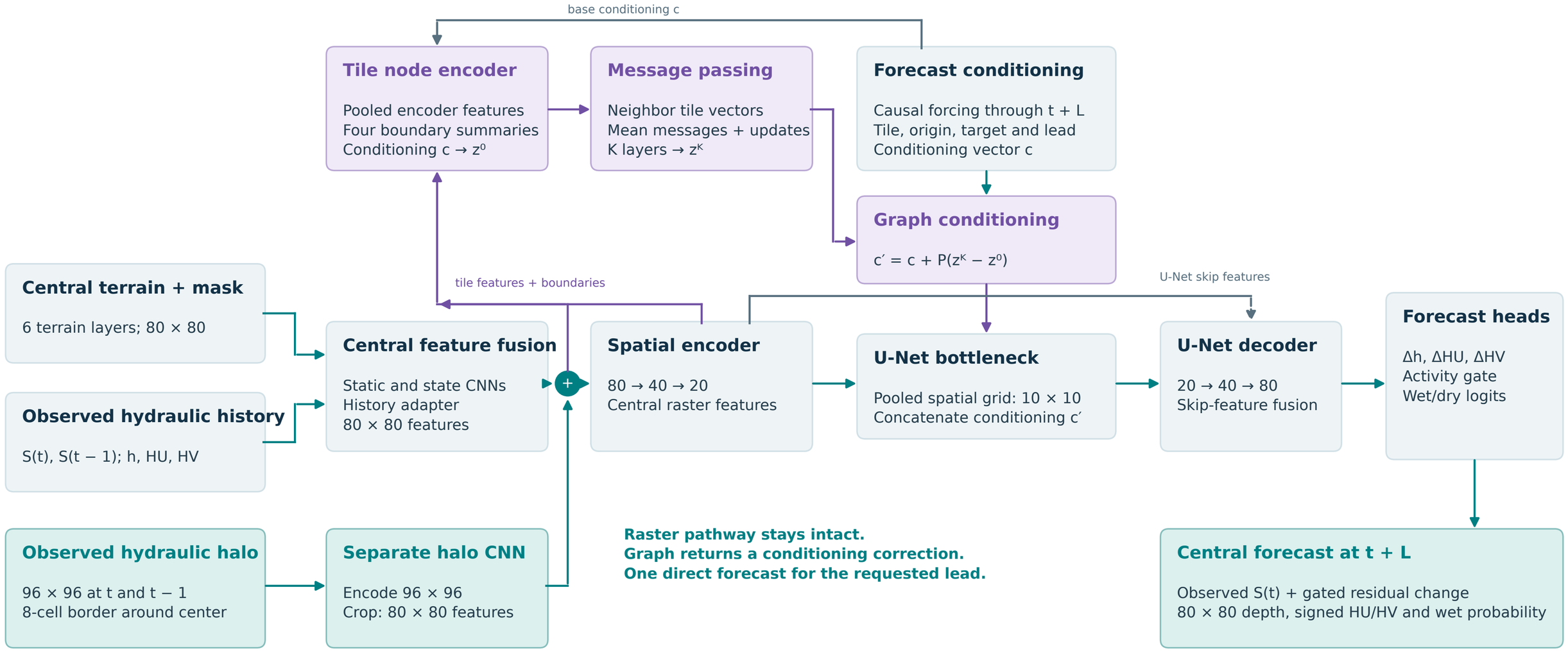
Direct U-Net with halo and tile graph context

Unified architecture | Halo enters early spatial features; optional graph messages modify bottleneck conditioning



Purple: graph context branch. Learned messages are features, not hydraulic fluxes; hierarchy is separate future work.

Implemented local graph prototype; corrected-data forecasting benefit remains unproven.